Technical report: Feature Engineering & Model Optimization Report

AI & ML Task 2: Comparative Analysis and Model Serialization

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1. Objective

The primary focus of this task was to transition from basic model training to a professional Machine Learning workflow. The project aimed to improve predictive accuracy by implementing data preprocessing techniques and evaluating multiple regression algorithms to identify the most robust solution for housing price estimation.

2. Methodology

The implementation followed a standard industry-aligned pipeline:

- **Data Preparation:** The California Housing dataset was loaded, and the target variable ("HousePrice") was separated from the eight input features.
- **Feature Scaling (Critical Step):** A **StandardScaler** was applied to normalize the feature space. This ensures that high-magnitude features (like Population) do not disproportionately influence the model compared to smaller-range features (like House Age), leading to more stable learning.
- **Validation Strategy:** The data was split using an 80/20 train-test ratio with a fixed random state to ensure reproducible results.

3. Algorithm Benchmarking

Instead of relying on a single model, three distinct regressors were trained and compared:

- **Linear Regression:** Used as the baseline model for initial performance benchmarking.
- **Ridge Regression:** Implemented with L2 regularization to minimize potential overfitting.
- **Decision Tree Regressor:** Utilized with a depth of 5 to capture complex non-linear relationships that linear models may miss.

4. Model Performance Comparison

Performance was measured using Root Mean Squared Error (RMSE) and the R^2 Score.

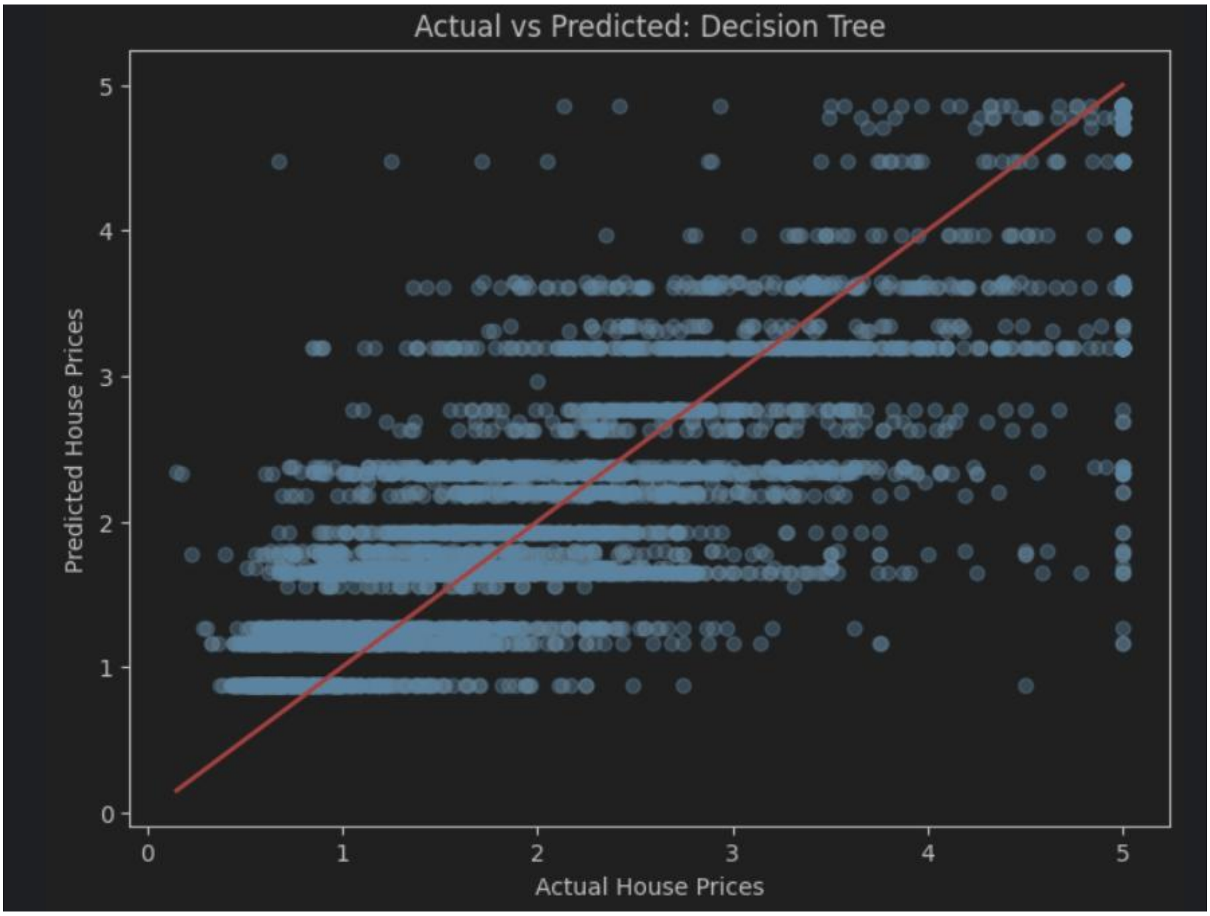
Model Algorithm	**RMSE (Accuracy)**	**R2 Score (Explanatory Power)**
Linear Regression	0.745581	0.575788

Model Algorithm	RMSE (Accuracy)	R2 Score (Explanatory Power)
Ridge Regression	0.745554	0.575819
Decision Tree (Selected)	**0.724234**	**0.599732**

Interpretation: A lower RMSE indicates higher prediction accuracy, while a higher R^2 score reflects better model fit.

5. Visual Validation

A scatter plot of **Actual vs. Predicted House Prices** was generated for the Decision Tree model. The close alignment of data points along the reference line confirms that the model successfully captured the underlying trends of the dataset.



"Actual vs Predicted: Decision Tree"

6. Final Conclusions & Selection

The **Decision Tree Regressor** was selected as the final model due to its superior R^2 score of **0.5997**. This model outperformed the linear baselines by successfully identifying non-linear patterns within the California housing data.

Model Persistence: As part of the advanced requirements, the best-performing model was successfully serialized and saved as **best_house_price_model.pkl** using the **joblib** library for future deployment.